Lesson Notes: Sensing Blocks and Operators in Scratch 3

1. Lesson Introduction

In Scratch, sprites can **interact with the environment** and make **decisions** based on conditions. This is possible using **Sensing blocks** and **Operators**.

- **Sensing blocks** help a sprite **detect things** (like touching, mouse position, or keyboard input).
- **Operators** help a sprite **calculate, compare, or combine values**.

Together, they allow sprites to respond intelligently and make **interactive projects**.

2. Learning Objectives

By the end of the lesson, learners should be able to:

1. Explain the purpose of **Sensing blocks**.
2. Explain the purpose of **Operators**.
3. Use sensing blocks to detect **keyboard input, mouse position, or touching other sprites**.
4. Use operators to perform **calculations, comparisons, and logic operations**.
5. Combine sensing and operators to make **interactive Scratch projects**.

3. Key Vocabulary

- **Sensing** – detecting events or values in Scratch.
- **Operator** – a block that performs calculations or logic checks.
- **Condition** – a question that is either true or false.
- **Boolean** – a true/false value.

4. Sensing Blocks

Sensing blocks are found in the **Sensing category**. They allow sprites to detect:

Examples:

1. **touching [sprite]?** – checks if the sprite is touching another sprite or the mouse pointer.
2. **key [space] pressed?** – checks if a key is pressed.
3. **mouse down?** – checks if the mouse is clicked.
4. **ask [question] and wait** – lets the sprite ask the user a question.
5. **answer** – gets the user’s answer.
6. **distance to [sprite]** – measures the distance to another sprite.
7. **loudness** – measures sound input from the microphone.

Example Use:

```
if <key right arrow pressed?> then
change x by 10
```

□ The sprite moves right **only if the right arrow key is pressed**.

5. Operators

Operators are found in the **Operators category**. They help sprites **perform math and logic operations**.

Types of Operator Blocks:

1. **Arithmetic**
   - `+` → addition
   - `-` → subtraction
   - `*` → multiplication
   - `/` → division

   Example: `set score to (score + 1)`
2. **Comparison**
   - = → equal to
   - < → less than
   - > → greater than

   Example: `if <score > 10> then say "Winner!"`
3. **Logic**
   - `and` → true if both are true
   - `or` → true if at least one is true
   - `not` → reverses true/false

   Example: `if <key space pressed? and touching Ball?> then ...`
4. **Random Numbers**
   - `pick random (1) to (10)`

   Example: Move a sprite to a **random position**.
5. **Joining Text**
   - Combine text for messages: `join "Hello " answer`

6. Combining Sensing and Operators

- Sensing blocks give values (true/false or numbers).
- Operators **process these values** to make decisions.

Example:

```
if <(score > 5) and <key space pressed?>> then
say "You win!"
```

- The sprite reacts **only if both conditions are true**.

7. Teacher Demonstration (Step-by-Step)

1. Add a sprite (e.g., Cat).
2. Add a Ball sprite as an obstacle.
3. Code example:

```
forever
if <touching Ball?> then
say "Ouch!" for 2 seconds
change color effect by 25
```

4. Add a **keyboard control**:

```
if <key right arrow pressed?> then
change x by 10
```

5. Use an operator:

```
if <(score > 10) or <touching Ball?>> then
say "Game Over"
```

8. Guided Practice for Learners

Example 1- Basic calculator

This one uses the ask block to add, subtract, multiply and divide.

Expressions and String Operators (Join, letter, & length of)

The trainer can use the following examples to illustrate the use of each blocks to the students.

Gives the integer portion of a decimal number.
For example: Floor of 10.3 = 10, Floor of 7.8 = 7

Gives remainder of a division.
For example: 10 mod 5 = 0, 12 mod 7 = 5.

Rounds the given decimal to the nearest integer.
For example: Round 6.6 = 7, Round 9.3 = 9.

Approach 2 (Strings)

In Scratch, a "string" is a sequence of characters. Characters can be letters, numbers, or even symbols.

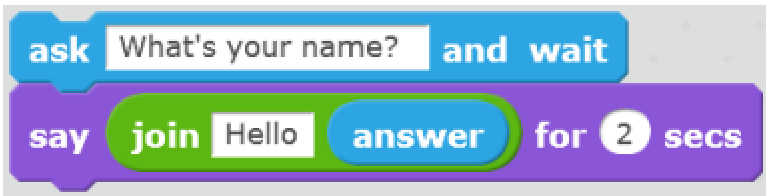
Examples:

For instance, Your name is a string of letters, or a sentence in a story is a string of words.

Concept

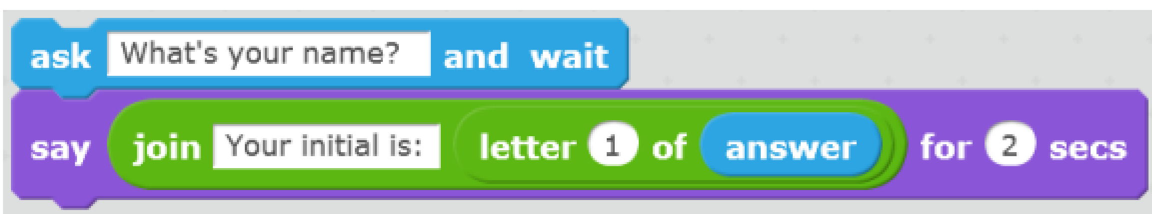
Join blocks, letter and the length of blocks and how to use each of them.

The **join operator** allows you to concatenate two strings:



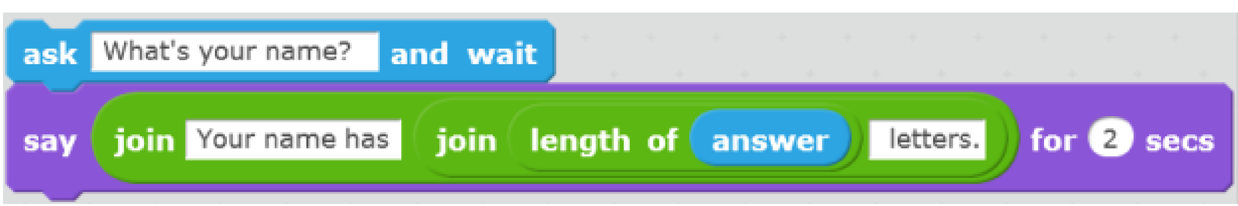
The above script asks you to type your name. If you type “George” the sprite will say “Hello George” on the screen.

The **“letter” operator** lets you get an individual letter of the given string:



The above script asks you to type your name. If you type “George” the sprite will say “Your initial is: G” on the screen.

The **“length” operator** tells you how many letters there are in the string:



The above script asks you to type your name. If you type “George” the sprite will say “Your name has 6 letters”.

Project

Create a calculator that displays the output using the join block.

Logical Operators

Approach

1.True/False Concept:

Begin with the idea of "true" and "false." For example, ask questions like:

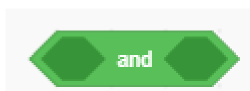
- Is it true that the sky is pink?
- Is it false that a cat can bark?

2. Introducing Logical Operators:

"and," "or," and "not" using simple examples:

The AND operator

AND: Shows that something can be true only if two conditions are true**.**



The AND operator

Condition 1	Condition 2	Combined effect (output)
False	False	False
False	True	False
True	False	False
True	True	True

Example 1

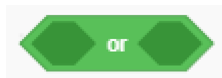
How the truth table works.

Explanation:

The code checks if both the space key and the right arrow key are pressed at the same time using the "and" operator.

If both conditions are true, the cat sprite will move to the right.

The OR operator



The OR operator

Condition 1	Condition 2	Combined effect (output)
False	False	False
False	True	True
True	False	True
True	True	True

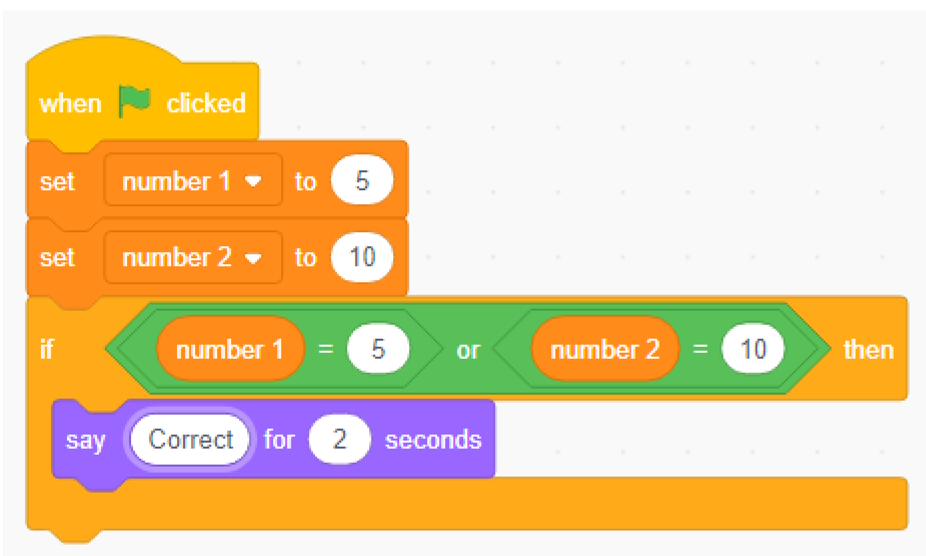
Example 2

In this example, the learners create two variable num 1 and num 2 and assign them values

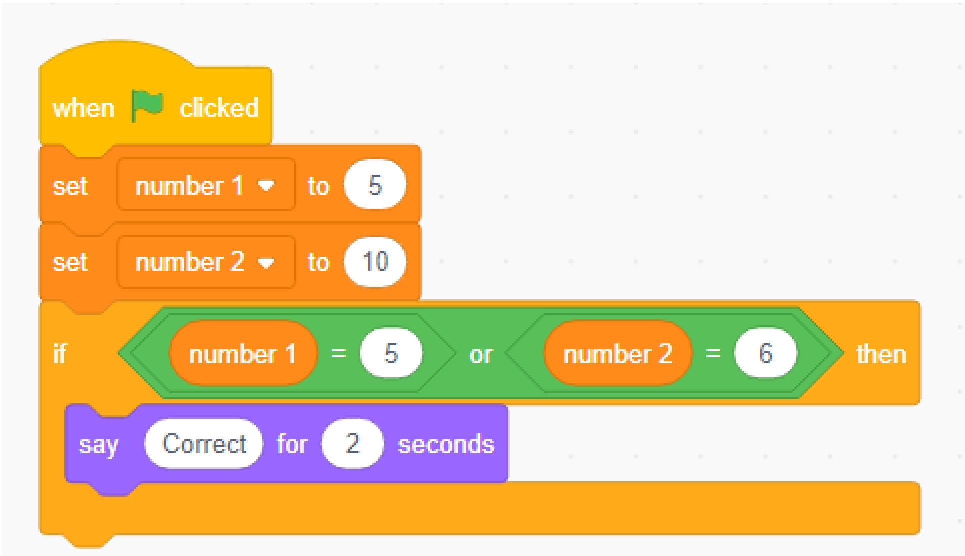
They then use the if block to check if the numbers are equal to the value written using the equal operator.

Since we are using the or operator, the condition inside the if statement will be executed if one of the statements is true or both are true.

Note: Always tap on the or operator to show the students whether its false or true.



For this code, the sprite shall say “correct” because both number 1 and number 2 are equal to those values because they were set at the top.



For this code, the sprite shall say “correct” because both number 1 is equal to 5 and his makes the statement true because they were set at the top.

9. Assessment Questions

1. Name two sensing blocks and what they do.
2. Name two operators and their use.
3. How can you detect if a sprite touches another sprite?
4. How can you use operators to make a decision?
5. Give an example of combining sensing and operators.

10. Summary

- **Sensing blocks** detect events, positions, or inputs.
- **Operators** perform calculations, comparisons, and logic.
- Together, they allow sprites to **make decisions and interact**.
- Using sensing and operators, learners can create **games, quizzes, and animations** in Scratch.